Lecture 4

What do words mean?

The Sequence Ontology (SO)

The Gene Ontology (GO)

What analyses are about

In general, most bioinformatics-oriented analyses results fall into two categories:

1. **What a piece of DNA is**: annotation or classification
2. **What a piece of DNA does**: functional analysis

And combinations of the two.

If you do research think about which category does your analysis fall into.

Ask someone to define "gene".

Now ask someone else.

It won't be the same definition.

Most scientists use the word "gene" to describe radically different "things".

Biology has many special words

Take the SGD_features.tab file for yeast annotations:

```
wget http://downloads.yeastgenome.org/curation/chromosomal
```

The second column of the file contains the *type* :

```
cat SGD_features.tab | cut -f 2 | sort | uniq
```

produces words like:

```
ARS
ARS_consensus_sequence
...
X_element_combinatorial_repeat
```

But what do `ARS`, `CDS` stand for?

Ontology

A structured vocabulary that offers a **definition** and a **classification** (taxonomy) of words.

Intended to remove ambiguity in the terminology

There could be multiple ontologies describing the same domain of knowledge from different perspectives.

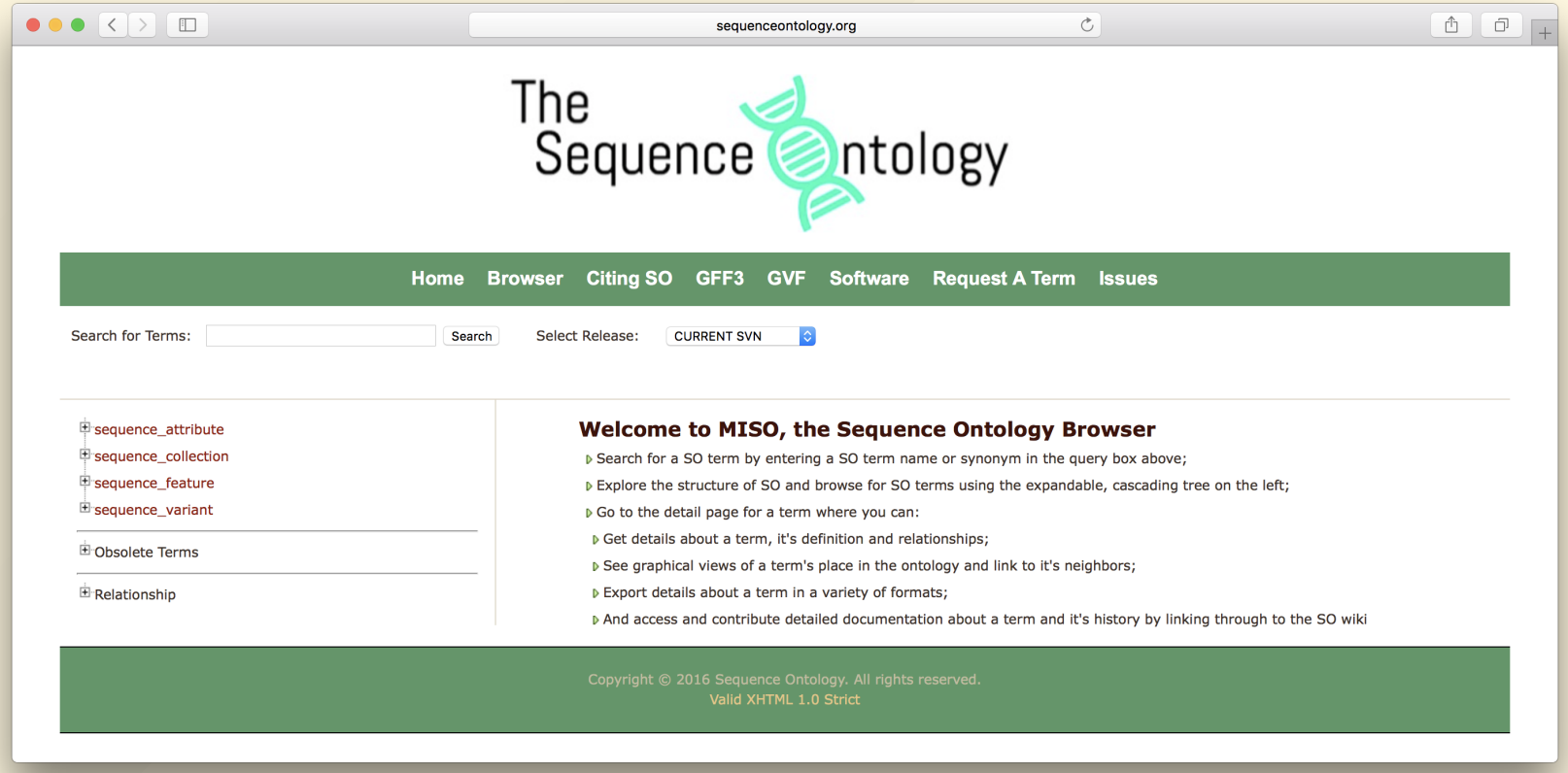
Biological ontologies

In this course we will deal mainly with two types of ontologies:

- **Sequence Ontology (SO)** deals with the definition of biological terms: what is a *gene*, what is a *transcript*. Is a *transcript* part of a *gene*?
- **Gene Ontology (GO)** deals with the functional characterization of genes. How many *different* functions are there? Which functions are *similar*? How do we group functions into *classes*?

The Sequence Ontology

The Sequence Ontology (SO) defines sequence features used in annotation.



So what is an `X_element_combinatorial_repeat`?

Look it up in the browser

Search the [Sequence Ontology Browser]

X_element_combinatorial_repeat (CURRENT_SVN)	
SO Accession:	SO:0001484 (SOWiki)
Definition:	An X element combinatorial repeat is a repeat region located between the X element and the telomere or adjacent Y' element.
Synonyms:	X element combinatorial repeat
DB Xrefs:	URL: http://www.yeastgenome.org/help/glossary.html
Parents:	repeat_region (SO:0000657)
	telomere (SO:0000624)

> *An X element combinatorial repeat is a repeat region located between the X element and the telomere or adjacent Y' element.*

The definition may have to be "defined" again.

The definition may contain other terms that you may not know:

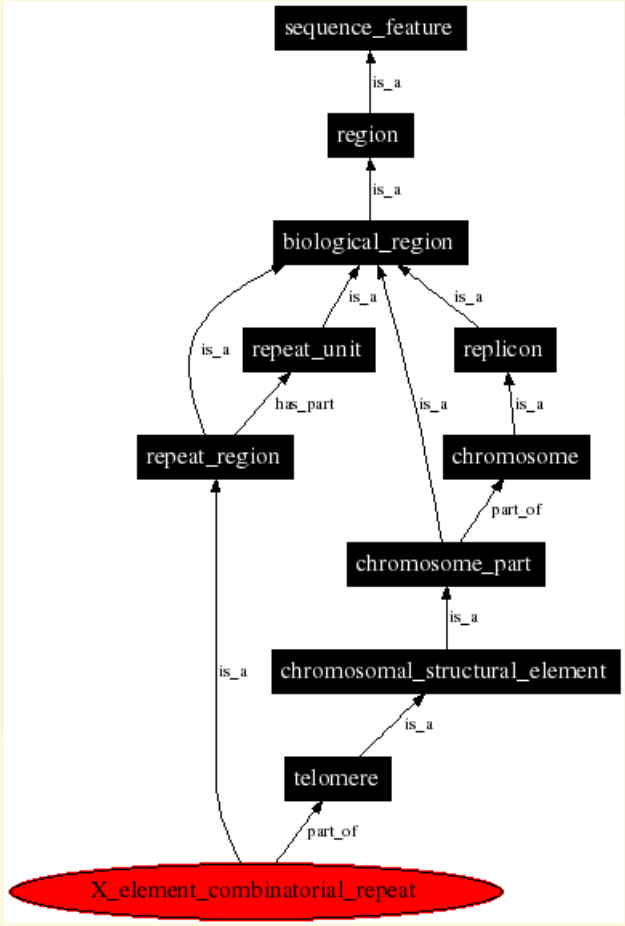
> *An `X element combinatorial repeat` is a `repeat region` located between the `X element` and the `telomere` or adjacent `Y' element`.*

So what is a:

- `repeat unit`, `X element`, `telomere`, `Y element`?

You can keep looking up each. The SO has definitions.

Graph (tree) representation



The ontology is also a hierarchical classification.

What is a "gene"?

The sequence ontology states:

> A region (or regions) that includes all of the sequence elements necessary to encode a functional transcript.
>
> A gene may include regulatory regions, transcribed regions and/or other functional sequence regions.

It is much broader concept than what most think.

What is the Gene Ontology (GO)?

The Gene Ontology (GO) is a controlled vocabulary that connects a gene product to one or more functions.

Calling it "Gene Ontology" is misleading.

GO categorizes gene products (proteins) rather than the genes themselves.

How is the GO designed?

The GO project has three independent sub-ontologies:

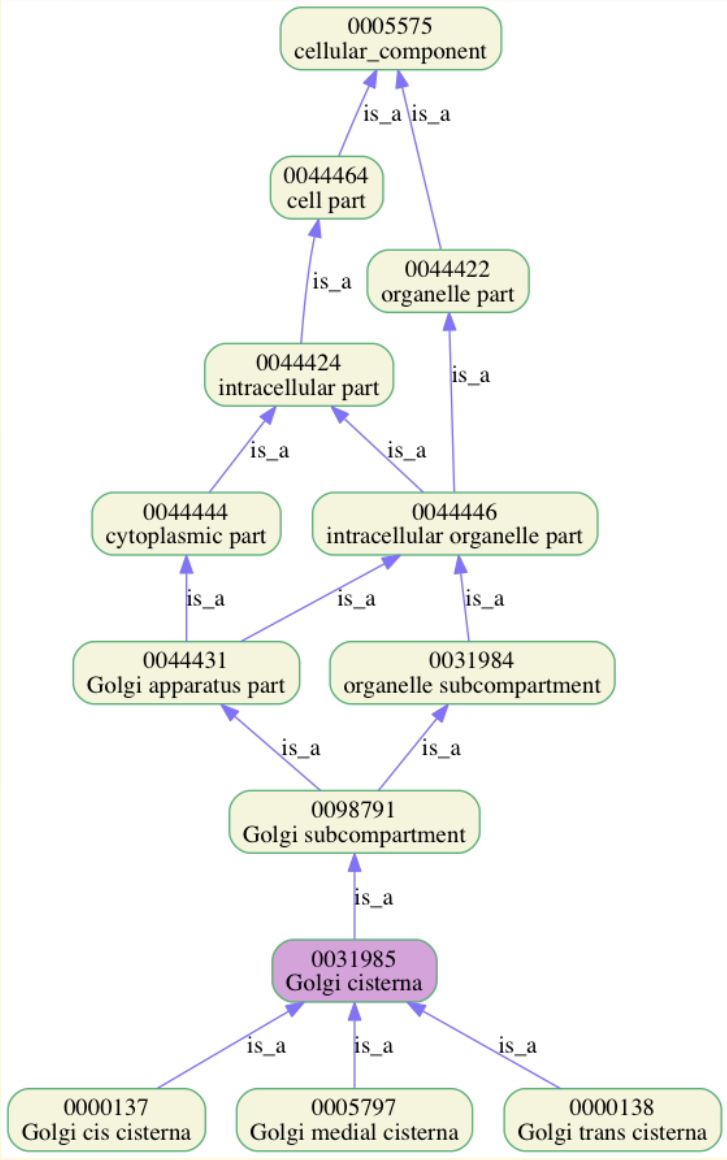
1. **Cellular component (CC).** Where does the product exhibit its effect? -> *cell, nucleus, Golgi membrane*
2. **Molecular function (MF).** How does it work at the molecular level? -> *lactase activity, actin binding*
3. **Biological process (BP).** What is the purpose of the gene product? Involves more than one distinct step: *transport, mitotic prophase, cholesterol efflux*

Where can the Gene Ontology be viewed?

The Gene Ontology website is the authoritative source for definitions, but is not particularly well suited for data interpretation.

The Quick GO service from the European Bioinformatics Institute offers a web interface with more user-friendly functionality.

What do GO relationships look like?



Association files

The first role of GO is to define functions.

The second role is to connect the functions to observed gene products.

The connections are called association files.

A gene product ID is connected to one or more GO functions.

Each organsims will have separate association files.

Gene Ontology (GO) summary

- The GO defines the words used to describe functions.
- The GO also stores the deposited knowledge on different organisms.
- The GO and the associations change over time.

The GO association files represent the accumulated knowledge of life sciences over many decades.

It is among the most essential components of life sciences! Yet most scientists know very little about it - or that it even exists.

Key concepts to remeber

For a typical analysis you need to use both.

First you need concepts from the **Sequence Ontology (SO)** â€“ What types of features are under study? How are the types interrelated?

Then you need concepts from the **Gene Ontology (GO)** â€“ What does a feature do? How does it do it? Where does it do it?